Calculate cuboid volume

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: Find the volume of a cuboid 12 cm by 7 cm by 5 cm. Copy or complete the first step.

2. A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

3. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the volume of a cuboid 12 cm by 7 cm by 5 cm. Copy or complete the first step.
   Answer 420 cm3.

2. A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer 6 cm.

3. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Both are 240 cm3.

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

Calculate cuboid volume

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

2. A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

3. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

4. Solve this independently and show every stage: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

5. Use a second method or representation for: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

6. Check this result using an inverse, estimate or known fact: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

7. Prove or explain your reasoning: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

Teacher answers and checking prompts

1. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
   Answer 420 cm3.

2. A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer 6 cm.

3. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Both are 240 cm3.

4. Solve this independently and show every stage: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
   Answer 420 cm3.

5. Use a second method or representation for: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer Expected result: 6 cm. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Expected result: Both are 240 cm3. A valid check must be shown.

7. Prove or explain your reasoning: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer 240 cubes. The explanation must show why the method works.

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Calculate cuboid volume

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ____________

1. Prove or explain your reasoning: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

2. Prove the answer to this question, rather than only calculating it: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

3. Solve this in two different ways and compare them: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

4. Work backwards from the answer to reconstruct the question: Both are 240 cm3.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.

6. Create a new example that tests the same idea as: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer 240 cubes. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
   Answer Expected result: 420 cm3. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer Expected result: 6 cm. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: Both are 240 cm3.
   Answer One valid reconstruction is: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

6. Create a new example that tests the same idea as: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.